

Solution Architecture

Project: PocketSmart AI – Your Smart Budget & Recommendation Assistant

1. Architecture Overview

- PocketSmart AI follows a web application architecture with a responsive frontend, FastAPI backend, AI recommendation layer, and data storage.
- The deployed application runs with Uvicorn and exposes API routes for the planner workflows.

2. Architecture Layers

Frontend Layer

- Responsive web pages for Home, Testimonials, Register, Login, Dashboard, planner pages, recommendation pages, and history.
- Forms collect user requirements and display AI-generated results.

Backend Layer

- FastAPI handles application requests and planner APIs.
- Authentication and session-related routes support user access.
- Uvicorn is used to run the FastAPI application.

AI Recommendation Layer

- Gemini 1.5 Flash Pro generates recommendation content from planner inputs.
- Separate planner endpoints process Home, Party, and Jewelry requests.

Data / History Layer

- Recommendation sessions are stored so users can access recommendation history and details.

3. Key API Routes

- /generate-home – Home planner recommendations.
- /generate-party – Party planner recommendations.
- /generate-jewelry – Jewelry planner recommendations.
- Authentication and session routes – user access and session handling.
- /recommendations-details – detailed recommendation view.
- /history – recommendation history.
- /startup – application startup handling.

4. Request Flow

- User enters planner requirements in the frontend.
- Frontend sends the request to the FastAPI backend.
- Backend processes the input and sends the relevant information to the AI recommendation layer.

- AI-generated recommendations are returned to the application.
- The result is displayed to the user and the session can be accessed through history.

5. Deployment

- The application is deployed as a web service and served using Uvicorn.
- The deployed system provides access to the PocketSmart AI planners and recommendation history.